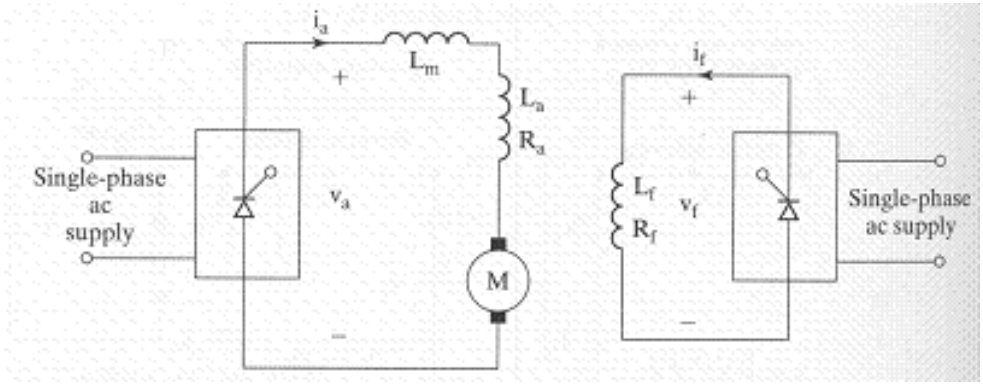
Electric Motors

Power semiconductor drive

Prepared by Dr Mahdi Alshamasin

Single-Phase Drives

- If the armature circuit of a dc motor is connected to the output of a single-phase controlled rectifier, the armature voltage can be varied by varying the delay angle of the converter α_a.
- The basic circuit agreement for a single-phase converter-fed separately excited motor is shown in Figure ( )
- At a low delay angle, the armature current may be discontinuous, and this would increase the losses in the motor.
- A smoothing inductor, L_m, is normally connected in series with the armature circuit to reduce the ripple current to an acceptable magnitude.
- A converter is also applied in the field circuit to control the field current by varying the delay angle α_f.



Single-Phase Drives

- Depending on the type of single-phase rectifiers,

  single-phase drives may be subdivided into:

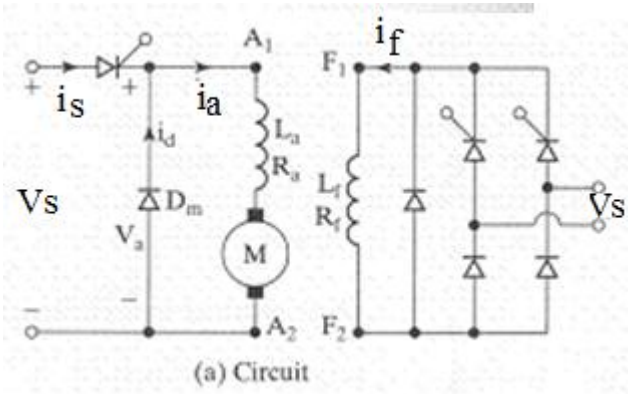
  – Single-phase half-wave-converter drives.

  – Single-phase semi converter drives (single-phase full-wave half-controlled converter drives)
  Single-phase full-converter drives (single-phase full-wave fully-controlled converter drives).

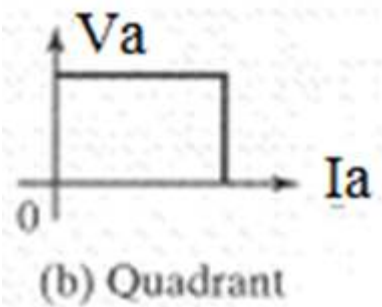
  – Single-phase dual-converter drives.

Single-Phase Half-Wave-Converter Drives

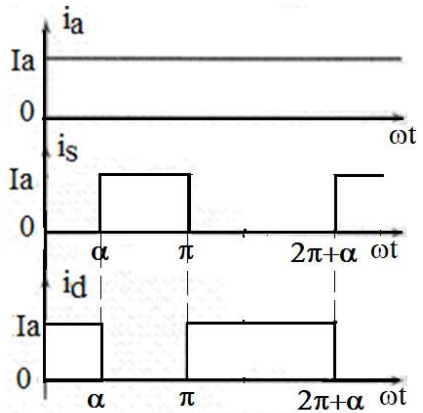
- A single-phase half-wave converter feeds a dc motor, as shown below.
- The armature current is normally discontinuous unless a very large inductor is connected in the armature circuit.
- A freewheeling diode is always required for a dc motor load and it is a one-quadrant drive.
- The applications of this drive are limited to the 0.5 kW power level.
- Figure (c) shows the waveforms for a highly inductive load.
- A half-wave converter in the field circuit would increase the magnetic losses of the motor due to a high ripple content on the field excitation current.



(a) Circuit



(b) Quadrant

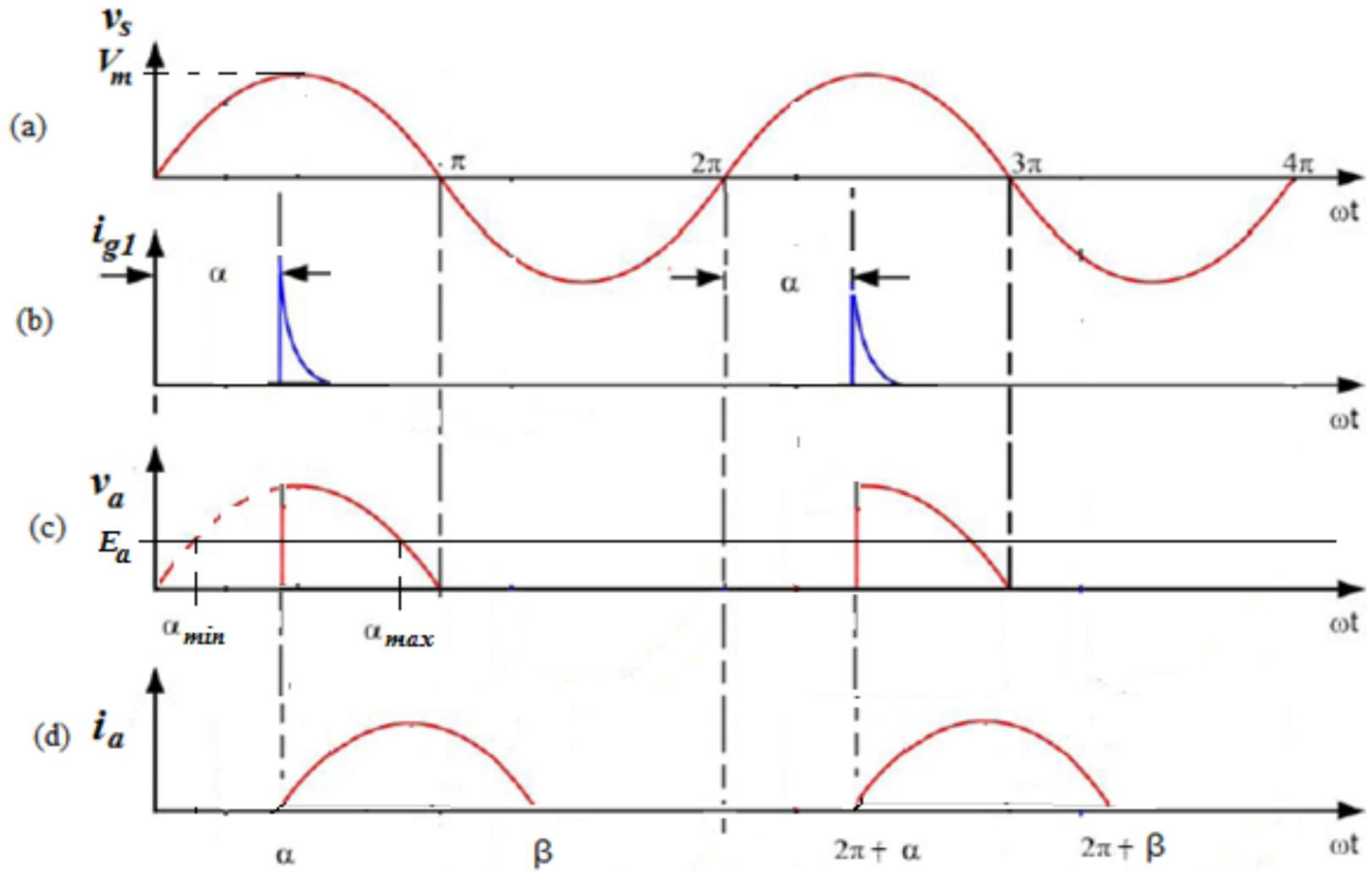


(c) waveforms

Single-Phase Half-Wave-Converter Drives

- Freewheeling diode connected across the motor terminals for the purpose of dissipation of energy stored in the inductance of the motor and to provide an alternative path for the motor current to allow the power switch to commutate easily.
- This type of drive is rarely used because of poor performance.
- Half-wave rectifier works in the first quadrant of the electromechanical characteristic with a ripple frequency equal to that of the source fr =fs , and the armature current in this rectifier is discontinuous.

Single-Phase Half-Wave-Converter Drives



α_{min} is the minimum firing angle below which the thyristor cannot be triggered. This angle can be calculated as,

$$V_m \sin \alpha_{min} = E_a \quad \rightarrow \quad \alpha_{min} = \sin^{-1}\left(\frac{E_a}{V_m}\right)$$

Similarly α_{max} is the maximum firing angle above which the thyristor cannot be triggered. Its value is given by

$$\alpha_{max} = \pi - \alpha_{min} = \pi - \sin^{-1}\left(\frac{E_a}{V_m}\right)$$

Single-Phase Half-Wave-Converter Drives

With a single-phase half-wave converter in the armature circuit, the armature voltage for Continuous conduction is

$$V_{a(av)} = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_s(\omega t)\, d\omega t = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin\omega t\, d\omega t = \frac{V_m}{2\pi}(1+cos\alpha) \quad \text{for} \quad 0 \le \alpha \le \pi$$

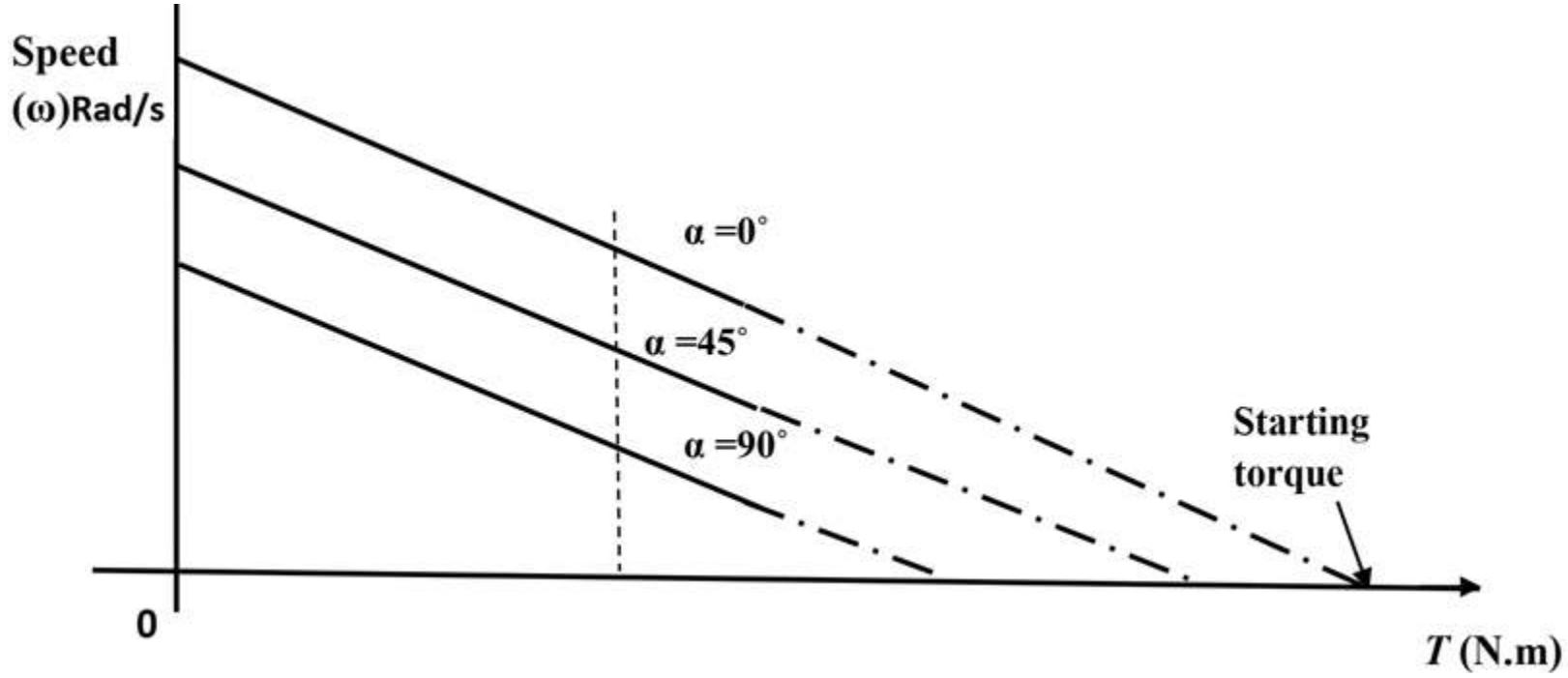
The motor angular velocity $\omega = \frac{V_m}{2\pi k}(1+cos\alpha) - \frac{R_a}{k^2}T$

where V_m is the amplitude of the supply voltage, $k=k_e\,\phi_n$

For the semiconductor in the field circuit, the average field voltage is

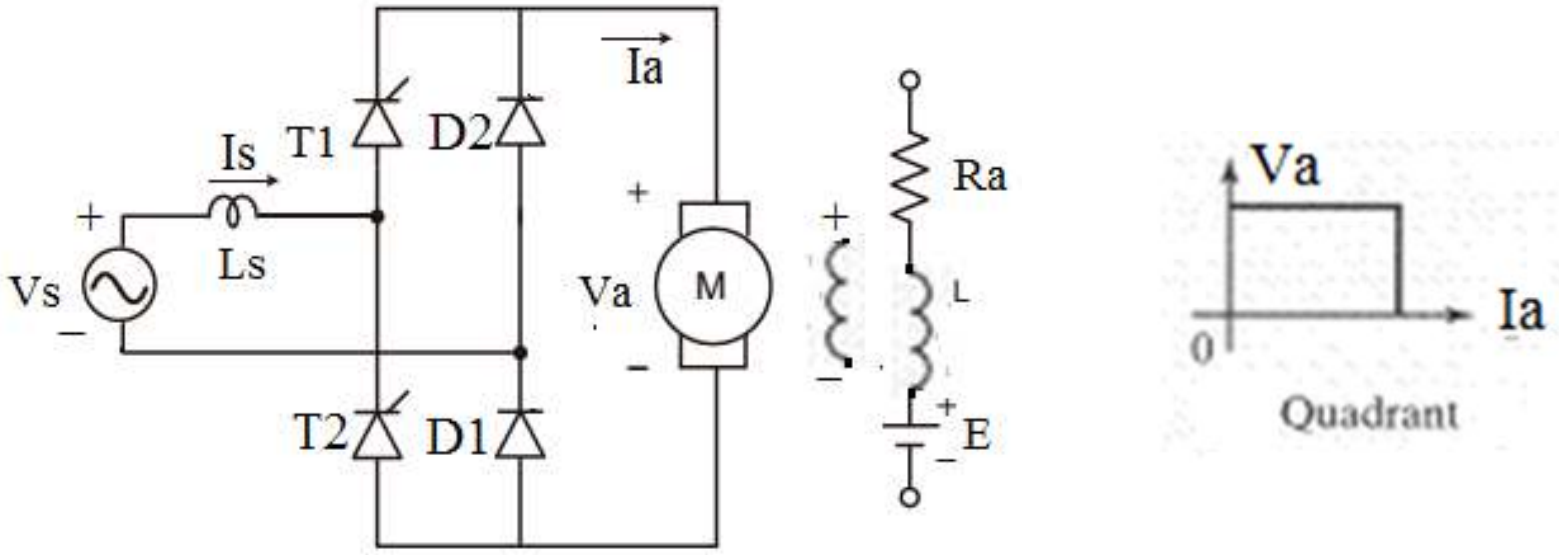
$$V = \frac{V_m}{\pi}(1+cos\alpha_f) \quad \text{for} \quad 0 \le \alpha_f \le \pi$$

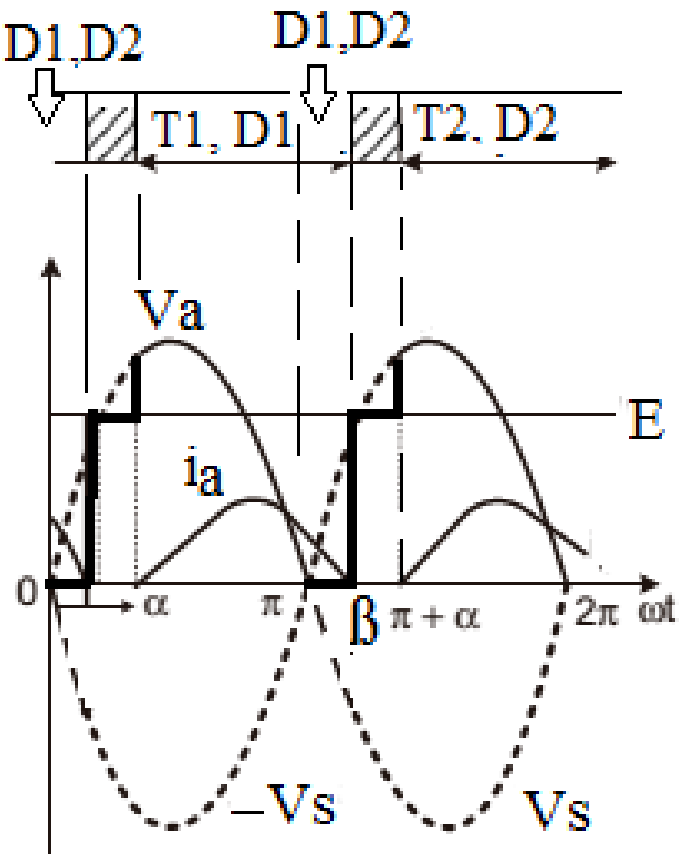
- Motor characacteristics



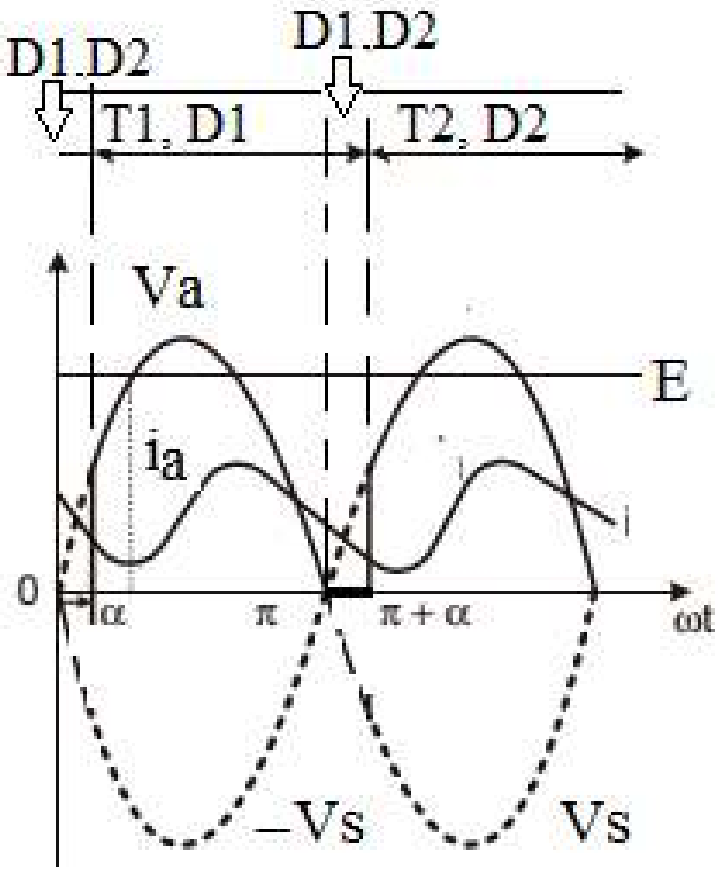
Single-phase semi converter drives(single-phase full-wave half-controlled converter drives)

- Semi converter is the converter that can be used for speed control consists of thyristors and diodes.
- The figure below shows the principal circuit and quadrant.
- the armature converter gives + V_a only, and allows operation in the first quadrant. Therefore, it is an one-quadrant drive.





discontinuous conduction



continuous conduction

Single-phase semi converter drives

For discontinuous conduction:

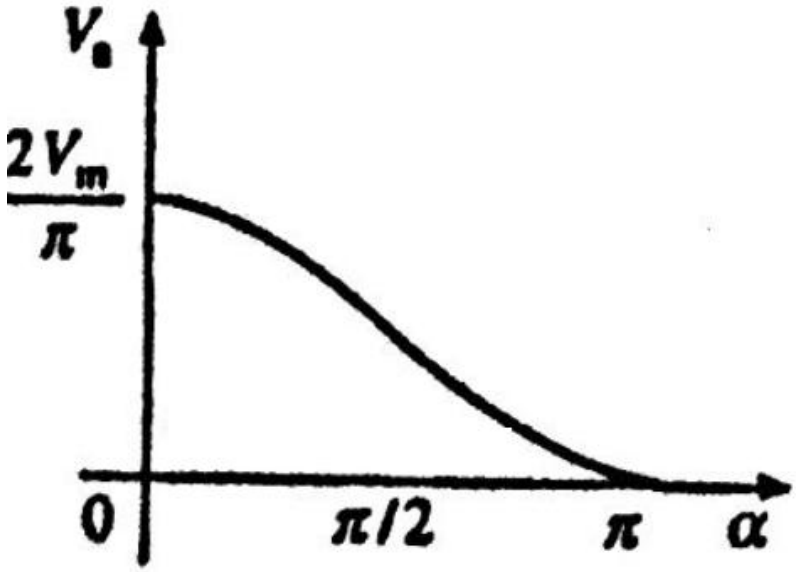
$$V_a = \frac{1}{\pi}\left[\int_{\alpha}^{\pi} V_m sin\omega t d(\omega t) + \int_{\beta}^{\pi+\alpha} E d(\omega t)\right] = \frac{V_m(1+cos\alpha)+(\pi+\alpha-\beta)E}{\pi}$$

$$\omega_m = \frac{V_m(1+cos\alpha)}{k(\beta-\alpha)} - \frac{\pi R_a}{k^2(\beta-\alpha)}T$$

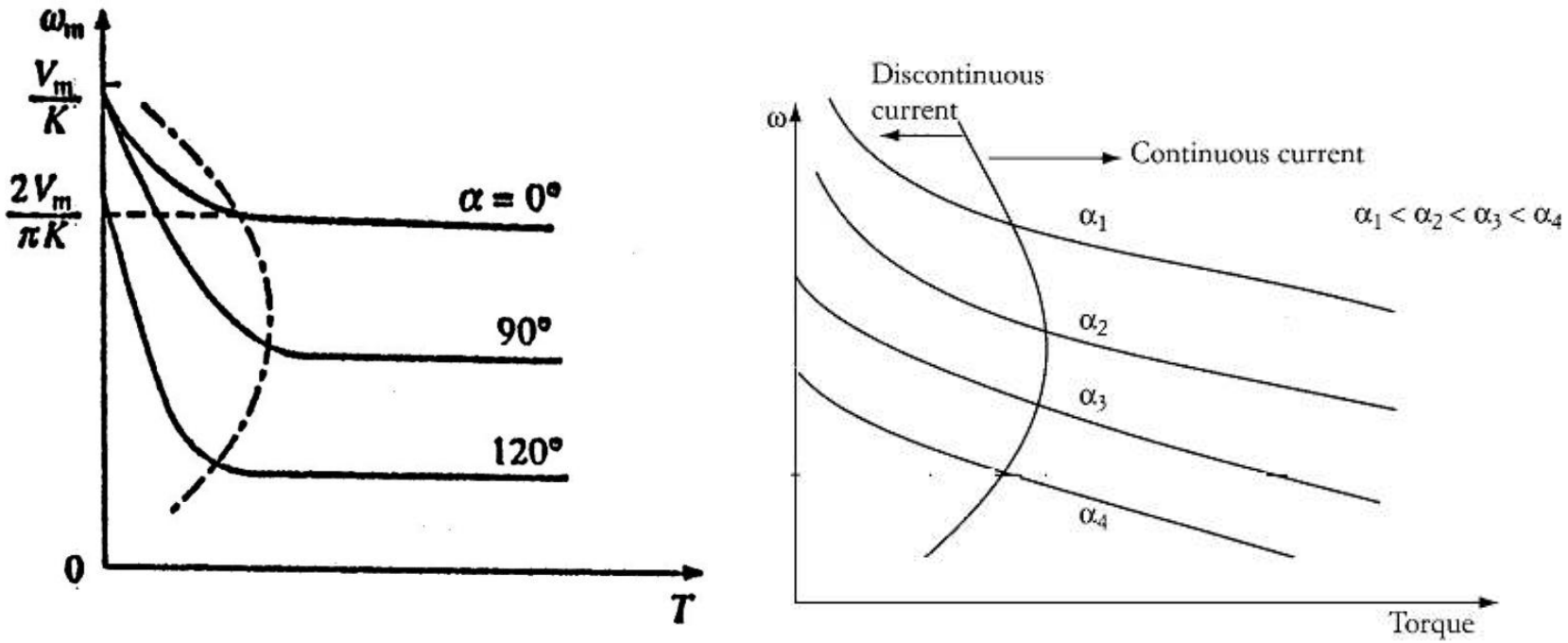
For continuous conduction:

$$V_a = \frac{1}{\pi}\left[\int_{\alpha}^{\pi} V_m sin\omega t d(\omega t)\right] = \frac{V_m}{\pi}(1+cos\alpha)$$

$$\omega_m = \frac{V_m}{\pi k}(1+cos\alpha) - \frac{R_a}{k^2}T$$

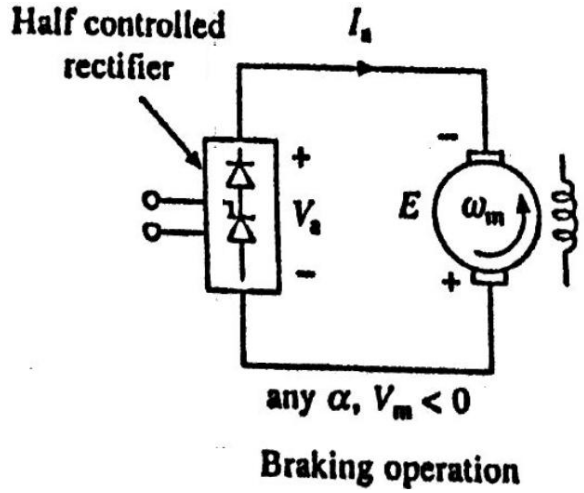


Single-phase semi converter drives



Speed torque carves of single-phase half-controlled rectifier fed separately excited dc motor

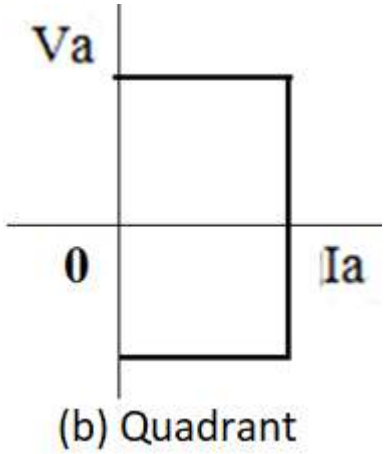
- The drive operates in the first quadrant only
- This drive shouldn’t be operated in IV quadrant
- This drive shouldn’t be used for active loads because the output voltage can’t be reversed. When the motor coupled to active load, the motor speed can reverse, reversing (E_a) as shown aside.
- As current direction doesn’t change, machine now works as a generator producing braking torque. Since rectifier voltage can’t reverse, generated energy can’t be transferred to ac source and therefore, it is absorbed in the armature circuit resistance.



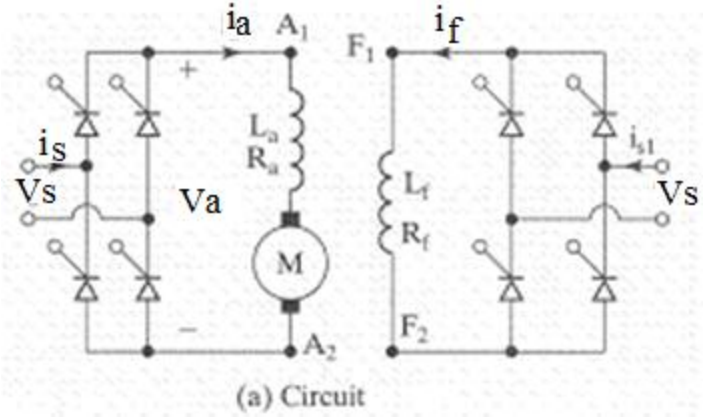
- Such a braking is not only inefficient but cause a a large current to flow through the rectifier and motor $I_a=(V_a+E_a)/R_a$.
- Since it can't be regulated by adjustment of firing angle, it will damage the rectifier and the motor.
- Therefore, when load is active, care should be taken to avoid such an operation. If such operation can't be avoided, fully control rectifier should be used.
- A half- controlled single-phase is cheaper and gives higher power factor compared to single-phase fully-controlled rectifier but then it only provides control in quadrant I.

Single-Phase Full-Wave-Converter Drives (single-phase full-wave fully-controlled converter drives)

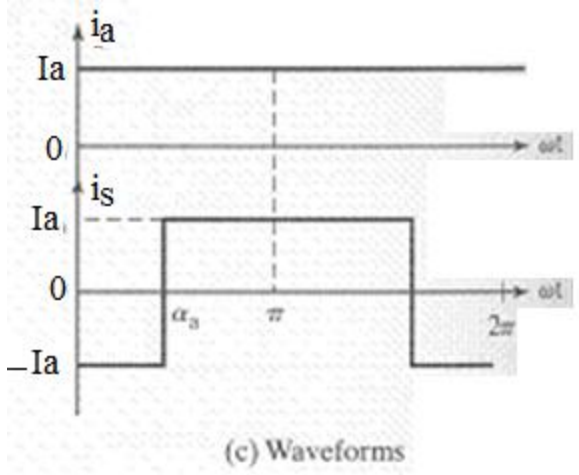
- The armature voltage is varied by a single-phase full-wave converter, as shown in Figure ( ).
- It is a two-quadrant drive, as shown in Figure ( ), and is limited to applications up to 15 kW.
- The armature converter gives + V_a or - V_a, and allows operation in the first and fourth quadrants.
- During regeneration for reversing the direction of power flow, the back emf of the motor can be reversed by reversing the field excitation.



(b) Quadrant



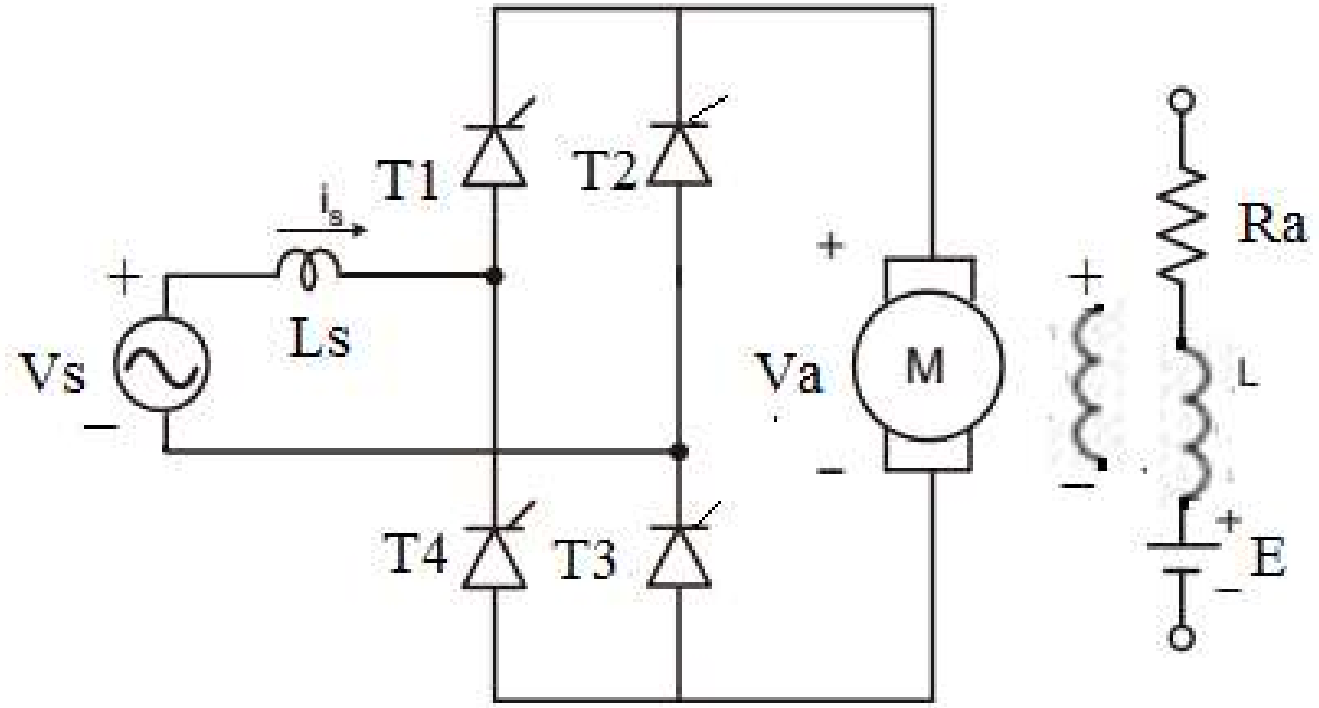
(a) Circuit

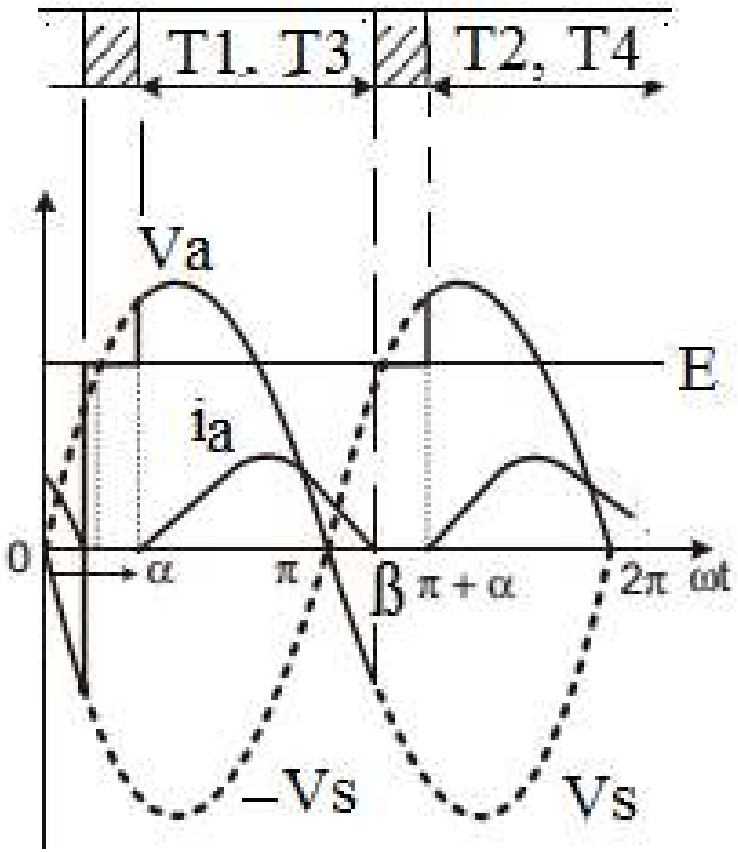


(c) Waveforms

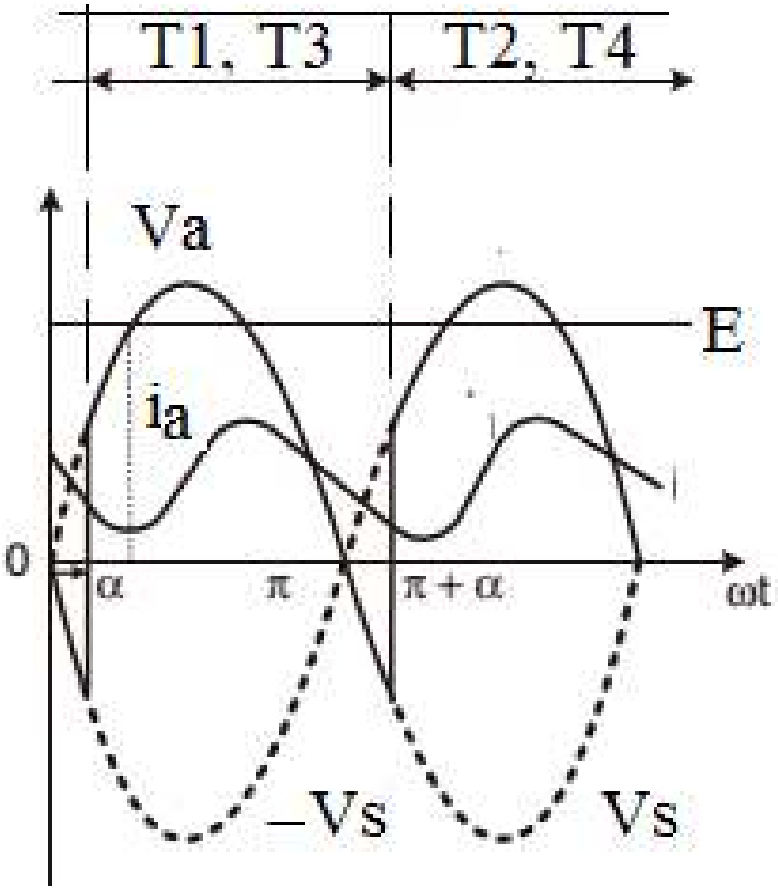
Single-Phase Full-Wave-Converter Drives

- The converter in the field circuit could be a full, or even a dual converter.
- The reversal of the armature or field allows operation in the second and third quadrants.
- The voltage and current waveforms for low and highly inductive load are shown in Figures below





Discontinuous conduction



Continuous conduction

Single-Phase Full-Wave-Converter Drives

Discontinuous condition

Drive equation is described by the following equations:

$$Va = R_a i_a + L_a \frac{di_a}{dt} + E = V_m sin\omega t \qquad \text{for} \quad \alpha \le \omega t \le \beta$$

$$Va = E, \quad i_a = 0 \quad \text{for} \quad \beta \le \omega t \le \pi + \alpha$$

Since the voltage drop across the armature inductance component is zero due to dc component of armature current , $Va = R_a i_a + E$,the average voltage is

$$Va = \frac{1}{\pi}\int_{\alpha}^{\beta} V_m \, sin\omega t d(\omega t) + \frac{1}{\pi}\int_{\beta}^{\pi+\alpha} E d(\omega t)$$

$$= \frac{V_m(cos\alpha - cos\beta) + (\pi + \alpha - \beta)E}{\pi}$$

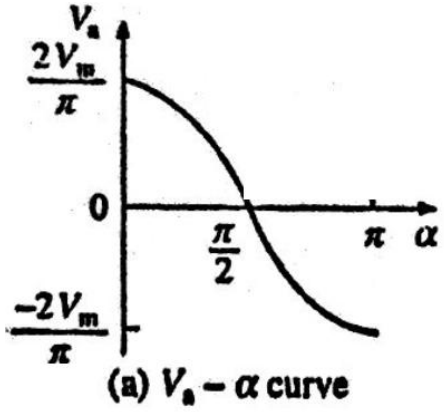
$$\omega_m = \frac{V_m(cos\alpha - cos\beta)}{k(\beta - \alpha)} - \frac{\pi R_a}{k^2(\beta - \alpha)} T$$

continuous condition

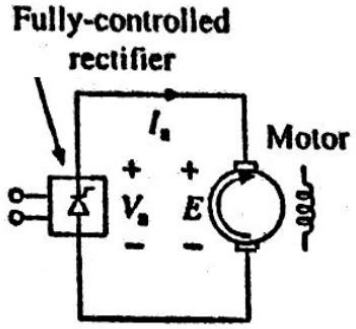
$$V_{a(av)} = \frac{1}{2\pi}\int_{0}^{2\pi} v_a(\omega t) d\omega t = \frac{2}{2\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin(\omega t)\, d\omega t = \frac{2V_m}{\pi}\cos\alpha$$

$$\omega_m = \frac{2V_m}{\pi k} cos\alpha - \frac{R_a}{k^2} T$$

Two-quadrant operation of the drive

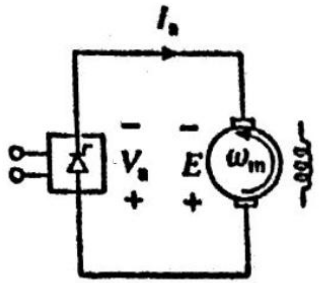


(a) $V_a - \alpha$ curve



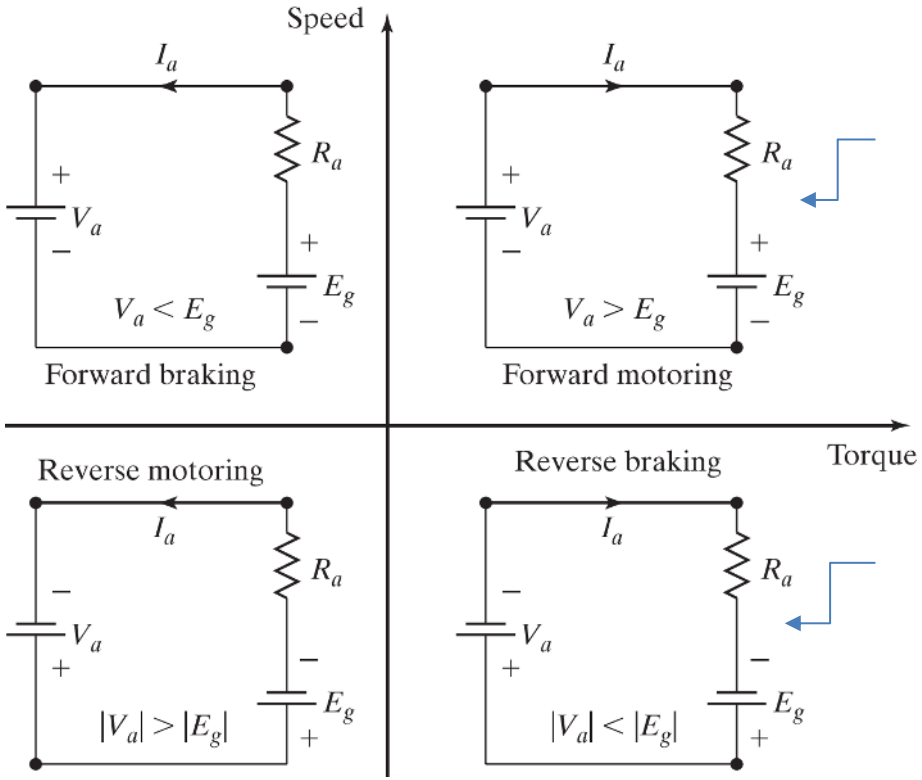
$\alpha < 90°$, $\omega_m > 0$

(b) Motoring

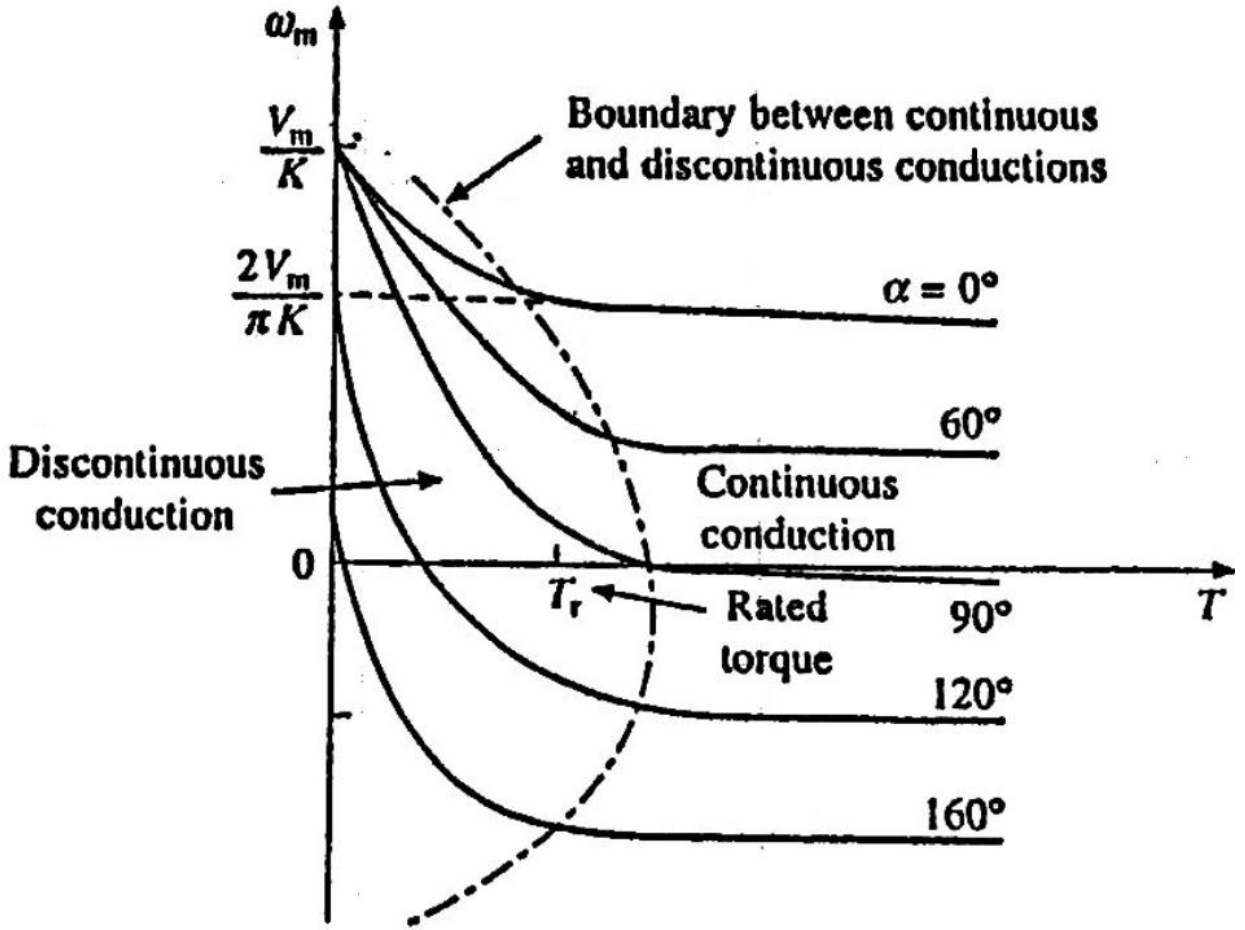


$\alpha > 90°$, $\omega_m < 0$

(c) Regenerative braking



Speed torque characteristics of Single-Phase Full-Wave-Converter Drives



Speed-torque characteristics of single-phase fully controlled rectifier fed dc separately excited motor

Speed –torque characteristics discussion

- Therefore, no-load speeds are given by

$$\omega_o = \frac{V_m}{k}, for\ 0 \leq \alpha \leq \pi/2$$

$$\omega_o = \frac{V_m sin\alpha}{k}, for\ \pi/2 \leq \alpha \leq \pi$$

- Maximum average terminal voltage ($2V_m/\pi$) is chosen equal to the rated motor voltage
- Ideal no-load speed of the motor when fed by a perfect direct voltage of rated value will then be ($2Vm/\pi k$), hence, the maximum no-load speed with rectifier control is ($\pi/2$) times this value.

- Boundary between continuous and discontinuous conduction is shown by doted line in the figure of ω=f(T).
- Boundary is reached when β=π+α at which the speed is equal to critical value of speed ω_{mc} which separates continuous conduction from discontinuous conduction for a given α
- For torques less than rated, a low power drive mainly operates in discontinuous conduction.
- In continuous conduction, the speed-torque characteristics are parallel straight line whose slope are depending on the armature circuit resistance

$$\omega_m = \frac{2V_m}{\pi k} cos\alpha - \frac{R_a}{k^2} T$$

Example

A 200V, 875 rpm, 150A separately excited dc motor has an armature resistance of 0.06 Ohm, it is fed from a single-phase fully- controlled rectifier with an ac source voltage of 220V, 50Hz. Assuming continuous conduction, calculate:

- Firing angle for rating motor torque and 750rpm.
- Firing angle for rating motor torque and (-500)rpm.
- Motor speed for $\alpha=160^{o}$ and rated torque.

Solution

At rated operation E=200-150x 0.06=191V.

- At 750rpm,

$$E = \frac{750}{875} \times 191 = 163.7V \qquad V_a = E + I_a R_a = 163.7 + 150 \times 0.06 = 172.7V$$

Now, $\frac{2V_m}{\pi} cos\alpha = V_a$ Or $\frac{2\times220\sqrt{2}}{\pi} cos\alpha = 172.7 \Rightarrow cos\alpha = 0.872 \Rightarrow \alpha = 29.3^{\circ}$

- At -500rpm,

$$E = \frac{-500}{875} \times 191 = -109V \qquad V_a = E + I_a R_a = -109 + 150 \times 0.06 = -100V$$

Now, $\frac{2V_m}{\pi} cos\alpha = V_a \Rightarrow \frac{2\times220\sqrt{2}}{\pi} cos\alpha = -100 \Rightarrow cos\alpha = -0.5 \Rightarrow \alpha = 120^{\circ}$

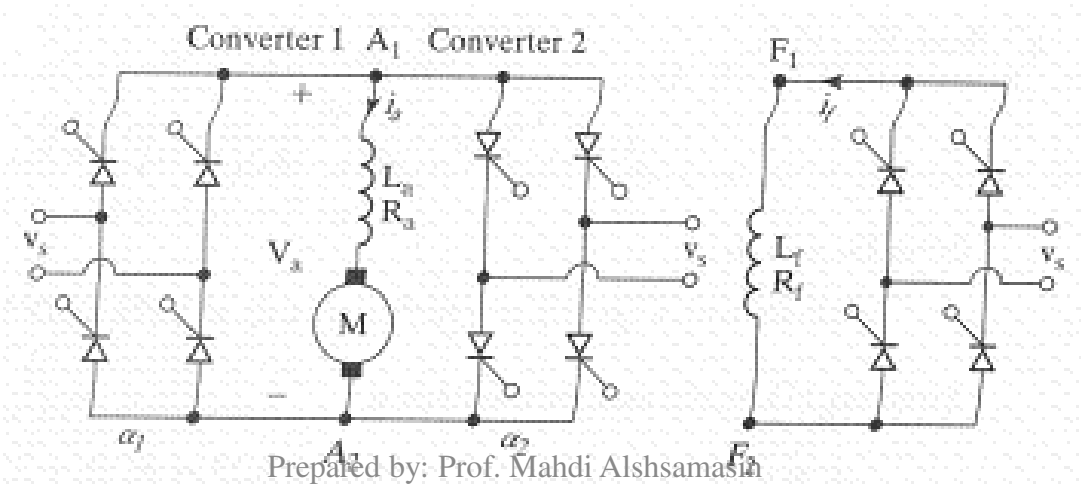
- At $\alpha = 160^{\circ}$

$$V_a = \frac{2V_m}{\pi} cos\alpha = \frac{2 \times 220\sqrt{2}}{\pi} cos160^{\circ} = -186V$$ Since

Va=E+IaRa, so, -186=E+150*0.06→E=-195V Speed $n = \frac{-195}{191} \times 875 = -893.2rpm$

Single-Phase Dual-Converter Drives

- Two single-phase full-wave converters are connected.
- Either converter 1 operates to supply a positive armature voltage, V_a, or converter 2 operates to supply a negative armature voltage, - V_a.
- Converter 1 provides operation in the first and fourth quadrants, and converter 2, in the second and third quadrants.
- It is a four-quadrant drive and permits four modes of operation: forward powering, forward braking (regeneration), reverse powering, and reverse braking (regeneration).
- It is limited to applications up to 15 kW. The field converter could be a full-wave or a dual converter.



Single-Phase Dual-Converter Drives

If converter 1 operates with firing angle α_1 , the average armature voltage is

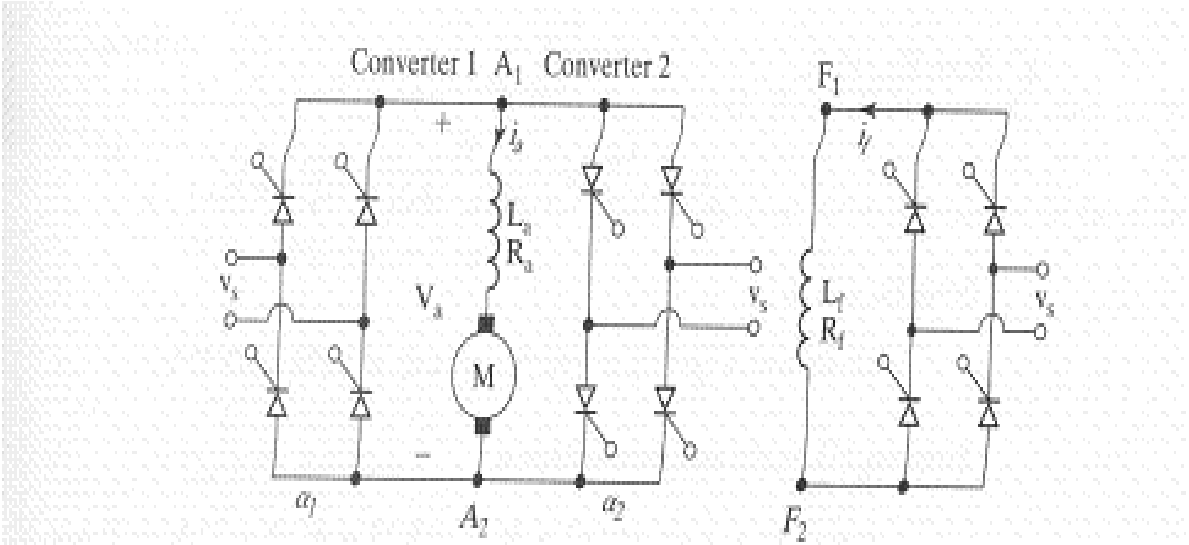
$$V_a = \frac{2V_m}{\pi}cos\alpha_1 \qquad \text{for} \qquad 0 \le \alpha_1 \le \pi$$

If converter 2 operates with firing angle α_2 , the average armature voltage is

$$V_a = \frac{2V_m}{\pi}cos\alpha_2 \qquad \text{for} \qquad 0 \le \alpha_2 \le \pi$$

With single-phase full wave converter in the field circuit, the field voltage is

$$V_f = \frac{2V_m}{\pi}cos\alpha_f \qquad \text{for} \quad 0 \le \alpha_f \le \pi$$



Three-Phase Drives

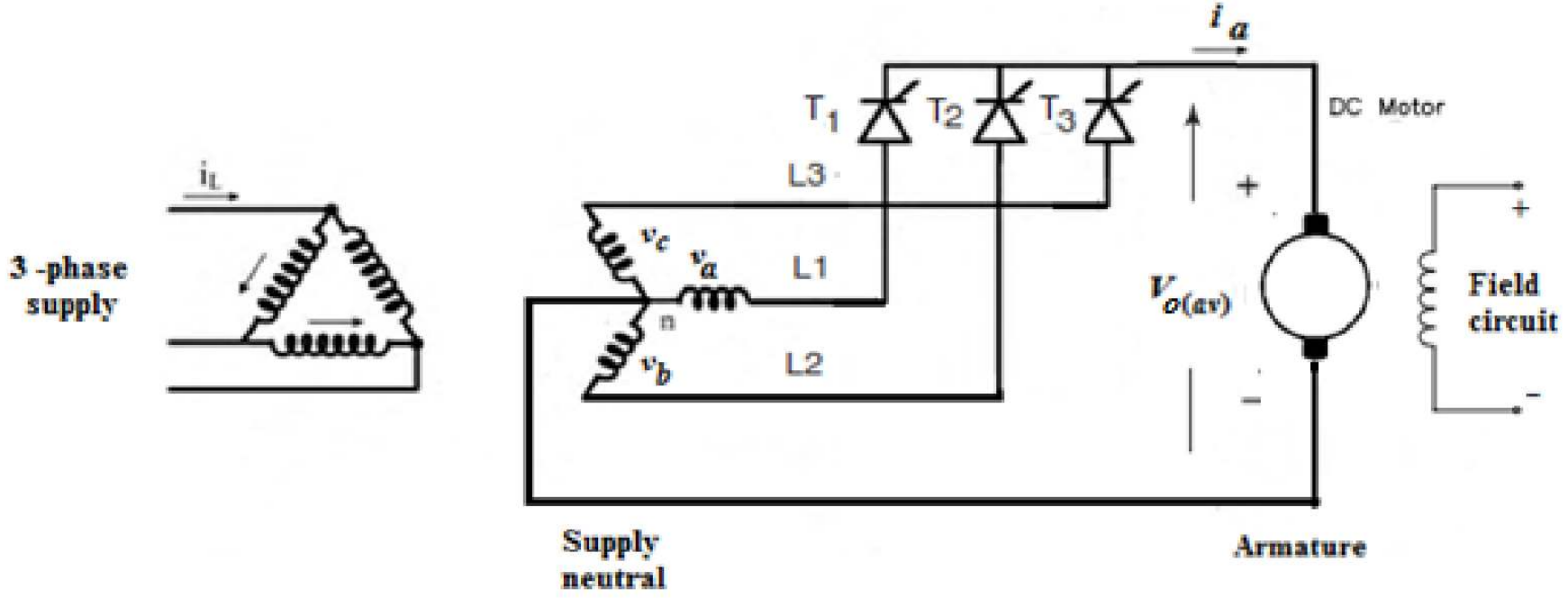
- The armature circuit is connected to the output of a three-phase controlled rectifier.
- Three-phase drives are used for high-power applications up to megawatt power levels.
- The ripple frequency of the armature voltage is higher than that of single-phase drives (fr=3fs or 6fs depending on the circuit) and it requires less inductance in the armature circuit to reduce the armature ripple current.
- The armature current is mostly continuous, and therefore the motor performance is better compared with that of single-phase drives.

Three-Phase Drives

- Similar to the single-phase drives, three-phase drives may also be subdivided into:
  - Three-phase half-wave-converter drives.
  - Three-phase semi converter drives (three-phase full-wave half-controlled converter drives).
  - Three-phase full-converter drives(three-phase full-wave fully-controlled converter drives).
  - Three-phase full-wave dual converter drives

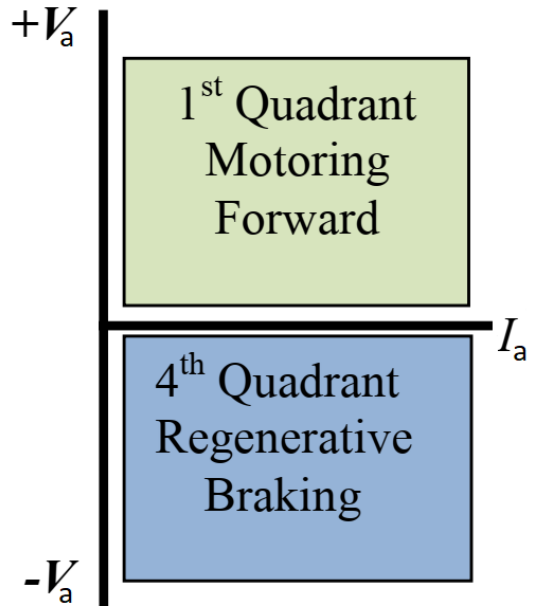
Three-Phase Half-Wave-Converter Drives

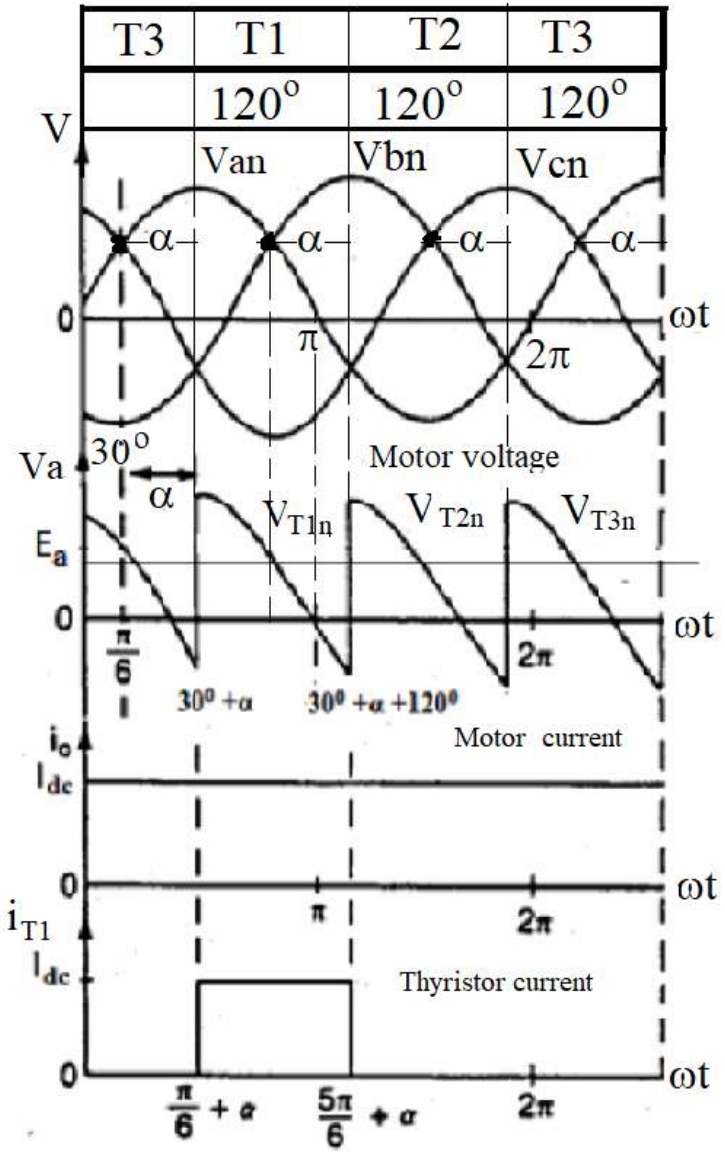
- A three-phase half-wave converter is only of theoretical importance and is generally not used in industrial applications because of the d.c. components inherent in its line currents. The dc motor fed by this converter operates in two quadrants. For medium size motors, in the range 15 – 120 kW, either the full-converters or semiconverter are used



Thyristors are fired in the sequence of their number with a phase difference of 120^0 by gate pulses of 120^0 duration.

-Each thyristor conducts for 120^0 applying respective line voltage to the motor.





Wave forms for the three-phase half-wave converter d.c. drive

Three-Phase Half-Wave-Converter Drives

To find the average value of the output d.c. voltage of the converter, let the transformer secondary phase to neutral voltage be,

$$V_{an} = V_m sin\omega t = \sqrt{2}V_{rms} sin\omega t$$

$$V_{bn} = V_m \sin(\omega t - \frac{2\pi}{3})$$

$$V_{cn} = V_m \sin(\omega t + \frac{2\pi}{3})$$

With three-phase half-wave converter in the armature circuit, if the converter operates with delay angle of α in continuous current conduction , the average armature voltage is

$$V_{o(av)} = \frac{1}{\frac{2\pi}{3}} \int_{30°+\alpha}^{30°+\alpha+120°} V_m \ sin\omega t \, d\omega t = \frac{3V_m}{2\pi} [-cos\omega t]_{30°+\alpha}^{150°+\alpha}$$

$$= -\frac{3V_m}{2\pi} [\cos(150° + \alpha) - \cos(30° + \alpha)]$$

$$= \frac{3\sqrt{3}V_m}{2\pi} cos\alpha$$

for $0 \leq \alpha \leq \pi$

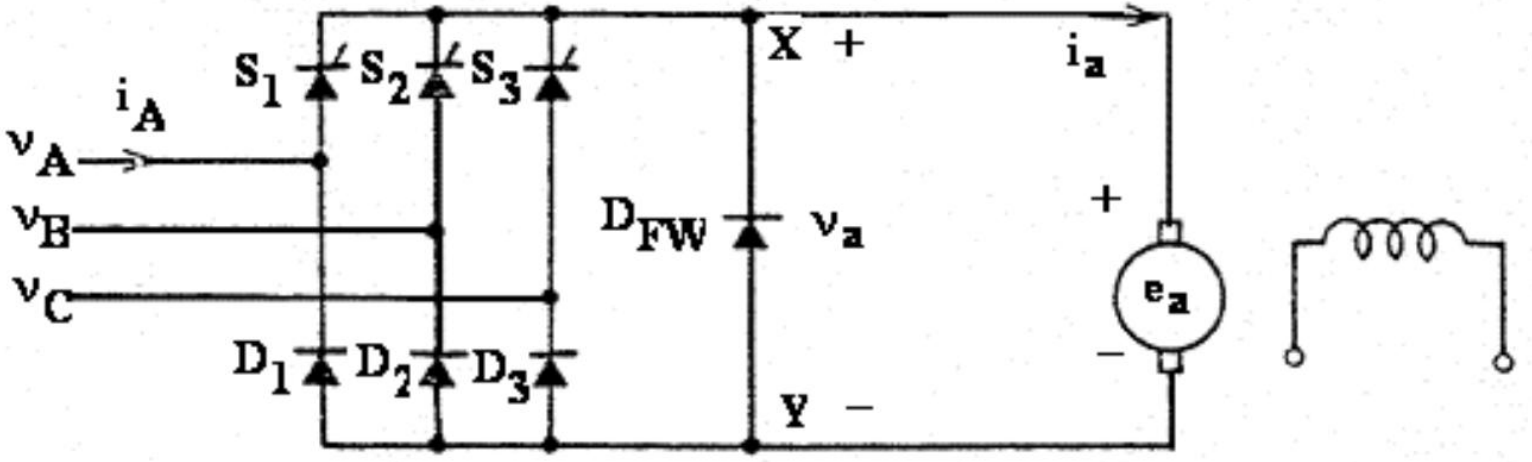
- Where V_m is a peak value of phase voltage of a Y- connected three-phase ac supply(peak of the supply line-neutral voltage).

- With three-phase semi converter in the field circuit, the field voltage is

$$V_f = \frac{3\sqrt{3}V_m}{2\pi}(1 + cos\alpha_f) \qquad \text{for} \qquad 0 \leq \alpha_f \leq \pi$$

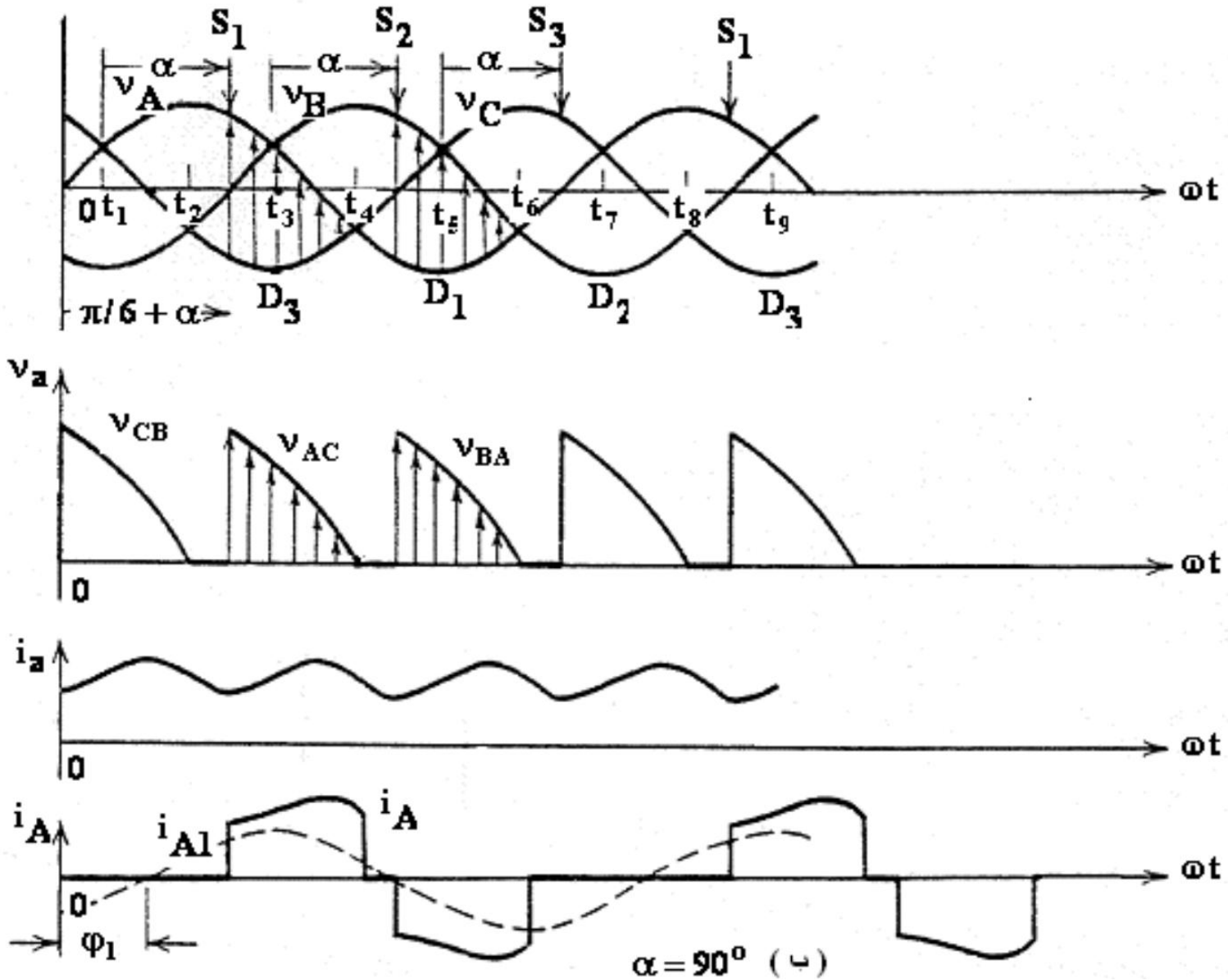
Three-phase semi converter (full-wave half-controlled converter) drives

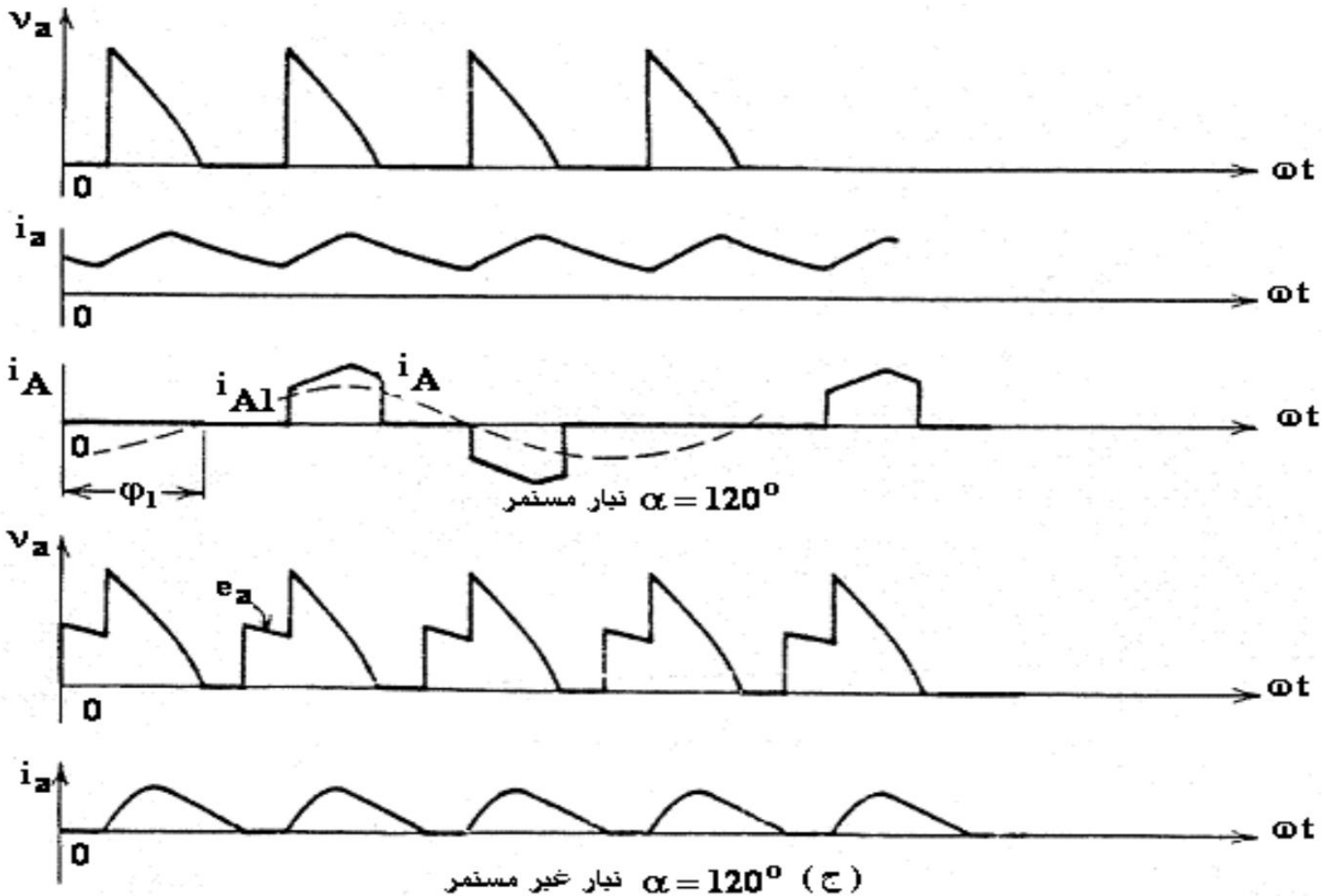
The three-phase semiconverter is a one-quadrant drive. Its circuit includes a freewheeling diode D_{FW} to maintaining continuous load current. It uses three thyristors and three diodes; hence a cost advantage is obtained compared with the full-converter. The circuit diagram for a separately-excited d.c. motor supplied from a three-phase a.c. supply through a three-phase semiconverter is shown in Fig. below



-Thyristors are fired in the sequence of their number with a phase difference of 120^0 by gate pulses of 120^0 duration.

-Each thyristor conducts for 120^0 and two switches (thyristor and diode) conduct at time, one from the upper group (thyristors) and the other from the lower group (diodes) applying respective line voltage to the motor.


S1
S2
S3
S1
α
α
α
νA
νB
νC
0 t1
t2
t3
t4
t5
t6
t7
t8
t9
ωt
π/6 + α
D3
D1
D2
D3
νa
νCB
νAC
νBA
0
ωt
ia
0
ωt
iA
iAl
iA
0
φ1
ωt
α = 90° (ب)


v_a
0
ωt
i_a
0
ωt
i_A
i_{A1}
i_A
0
φ_1
ωt
نبار مستمر $\alpha = 120^o$
v_a
e_a
0
ωt
i_a
0
ωt
نبار غير مستمر $\alpha = 120^o$ (ج)

Three-phase semi converter (full-wave half-controlled converter) drives

The instantaneous values of the source phase voltages can be written as

$$V_A = V_m sin\omega t = \sqrt{2}V_{rms} sin\omega t$$

$$V_B = V_m \sin(\omega t - \frac{2\pi}{3}) = \sqrt{2}V_{rms} sin(\omega t - \frac{2\pi}{3})$$

$$V_C = V_m \sin(\omega t + \frac{2\pi}{3}) = \sqrt{2}V_{rms} sin(\omega t + \frac{2\pi}{3})$$

Assuming continuous current operation, the average value of the armature voltage at the motor terminals is a contribution from the upper half-bridge plus a contribution from the uncontrolled lower bridge. Hence for all firing angles we can write:

$$V_{av} = \frac{3}{2\pi}\int_{\frac{\pi}{6}+\alpha}^{\frac{2\pi}{3}+\frac{\pi}{6}+\alpha}(V_A - V_C)d(\omega t) = \frac{3\sqrt{3}}{2\pi}V_m cos\alpha + \frac{3\sqrt{3}}{2\pi}V_m$$

$$= \frac{3\sqrt{3}}{2\pi}V_m(cos\alpha + 1)$$

Three-Phase Full-Wave-Converter Drives

- A three-phase full-wave-converter drive is a two-quadrant drive without any field reversal, and is limited to applications up to 1500 kW.
- During regeneration for reversing the direction of power,

However, the back emf of the motor is reversed by reversing the field excitation.

- The converter in the field circuit should be a single- or three-phase full converter.

For a three -phase full-wave converter in the armature circuit, the armature voltage is

$$Va = \frac{3}{\pi}\int_{\alpha+\frac{\pi}{3}}^{\alpha+\frac{2\pi}{3}} V_m \sin\omega t d(\omega t) = \frac{3V_m}{\pi}\cos\alpha$$

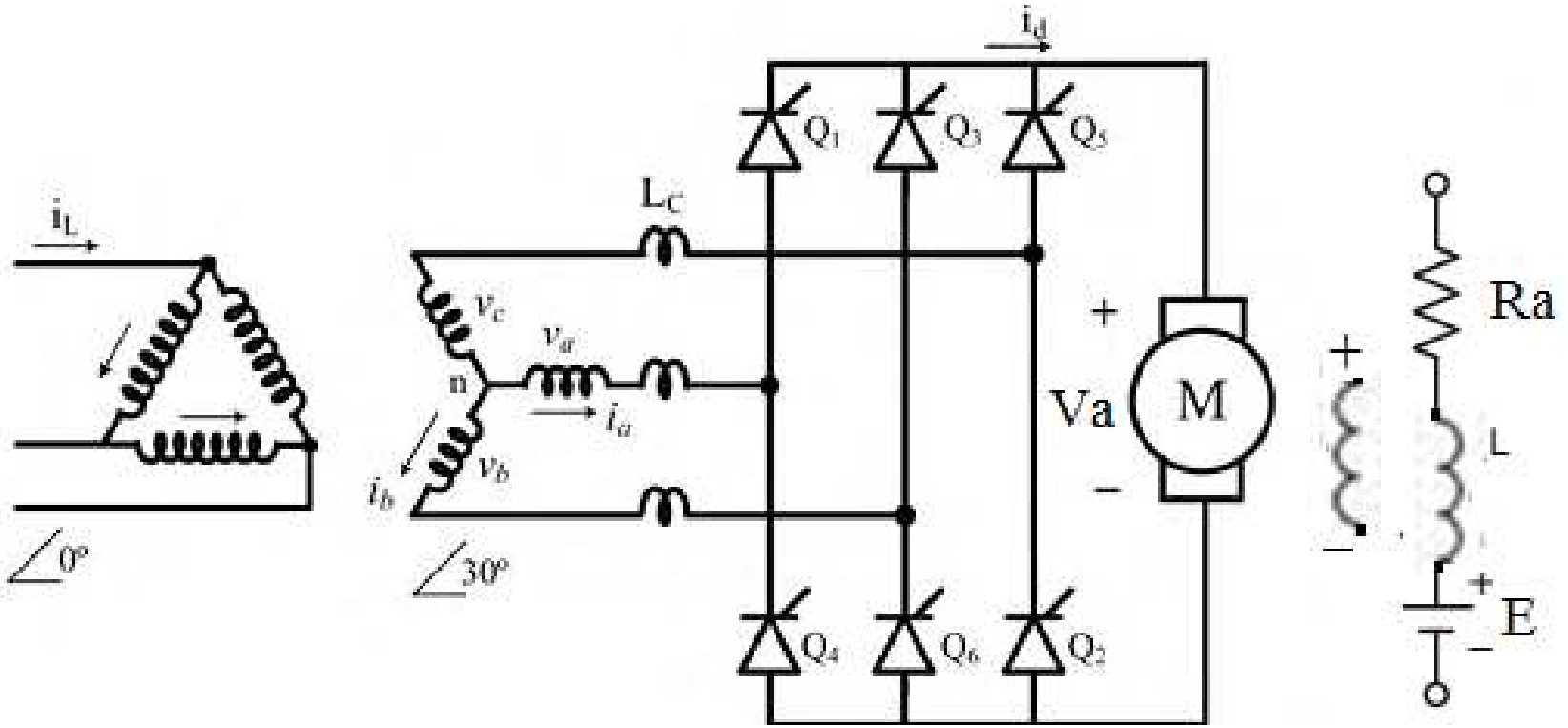
Where V_m is a peak value of line voltage of a Y- connected three-phase ac supply.

The motor speed is

$$\omega_m = \frac{3V_m}{\pi k}\cos\alpha - \frac{R_a}{k^2}T$$

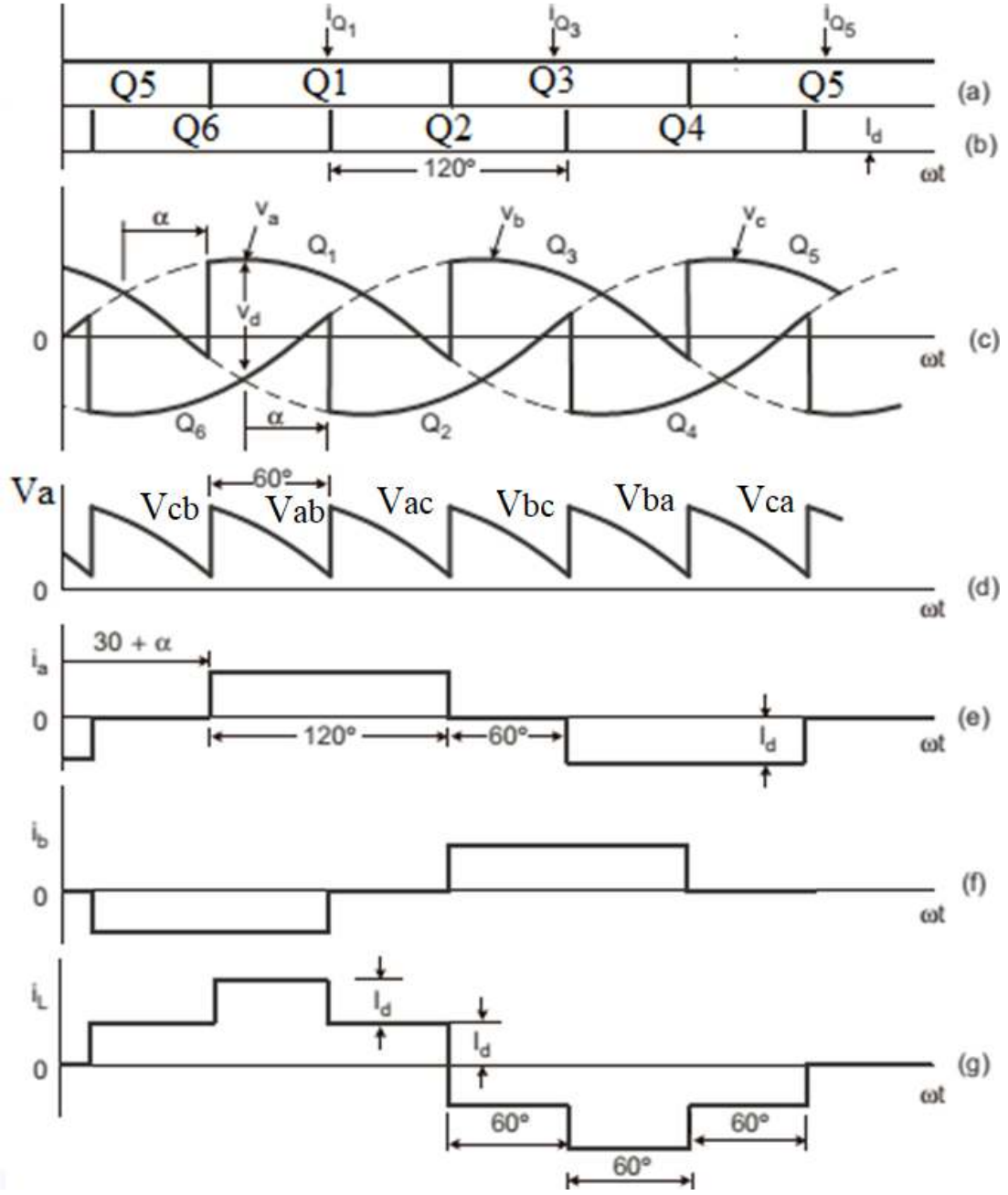
For a three -phase full-wave converter in the field circuit, the field voltage is $V_f = \frac{3V_m}{\pi}\cos\alpha_f$

Motoring mode at $\alpha = 30^o$

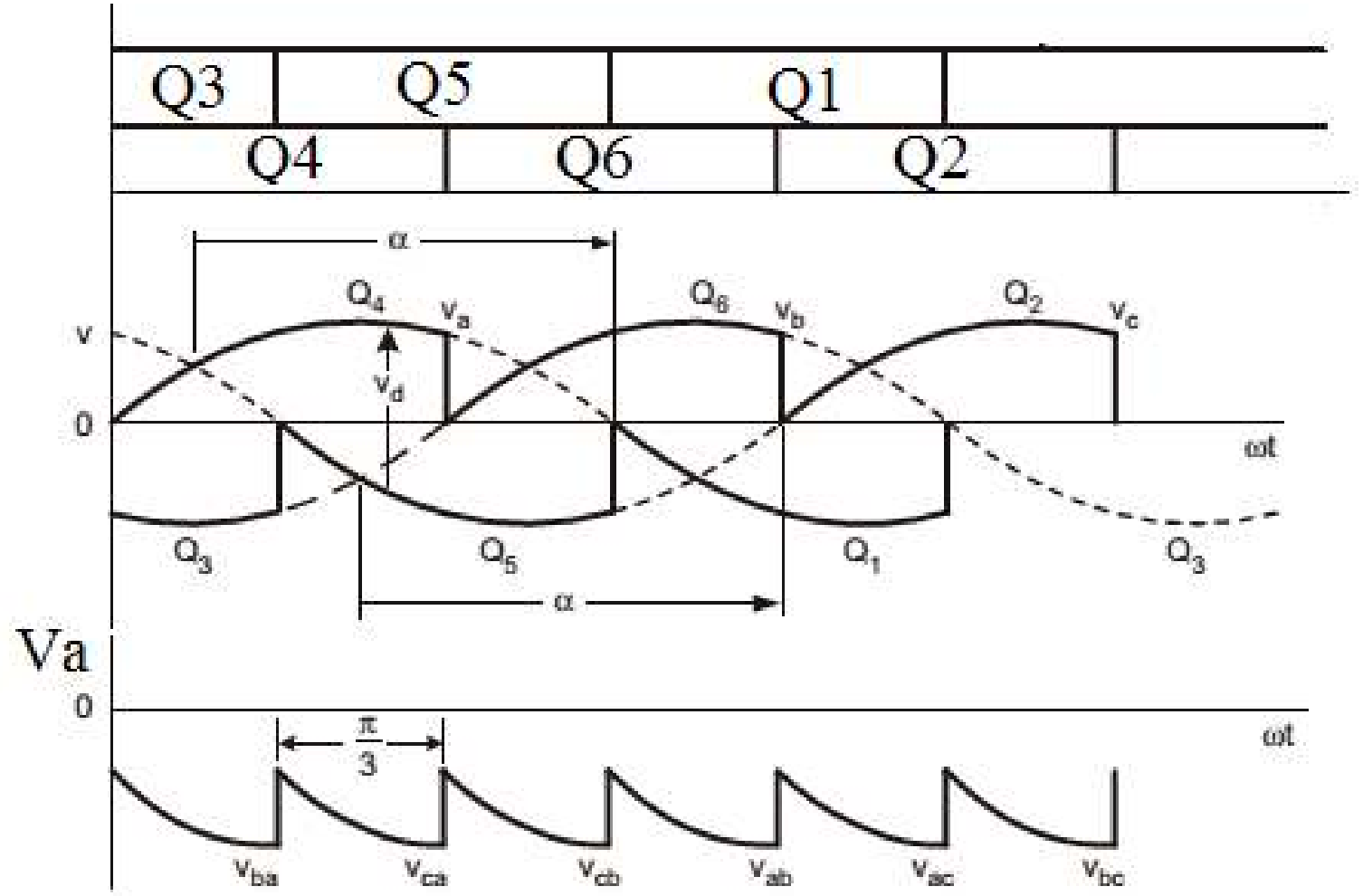


Motoring mode at

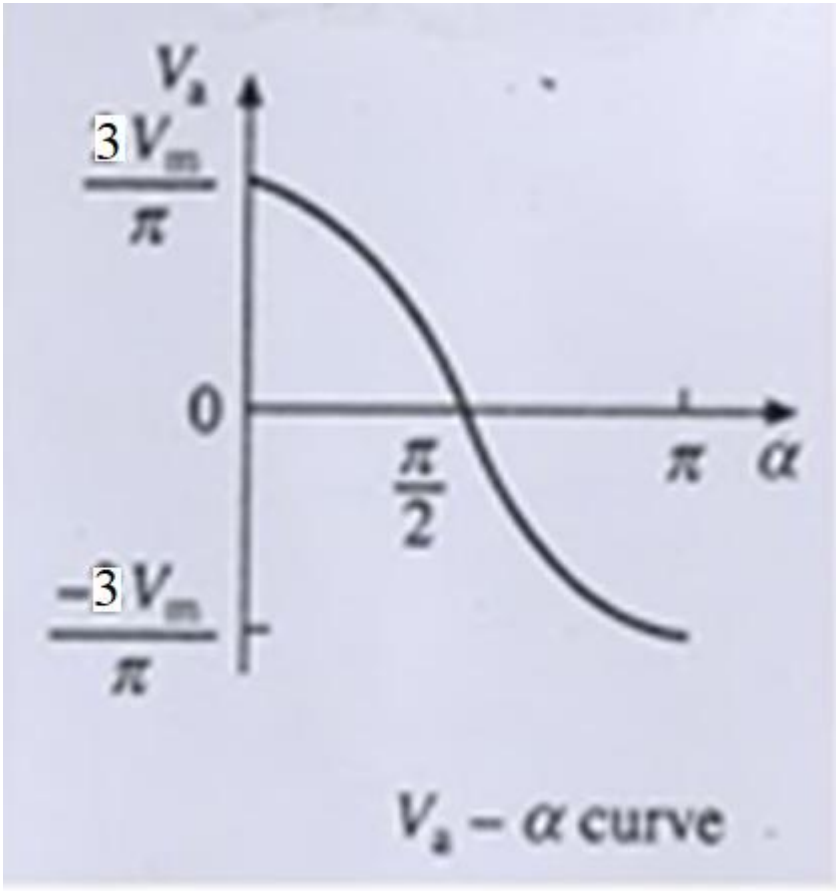
$\alpha = 30^0$

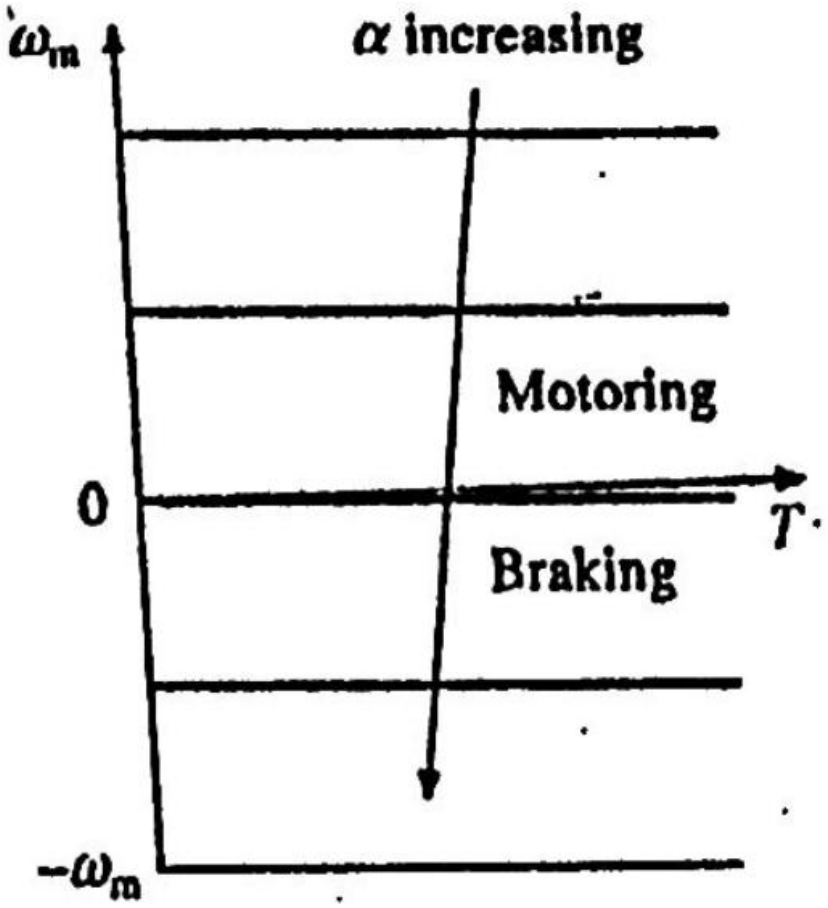


Braking (regenerative) mode at $\alpha = 150^0$



characteristics





discussion

-Thyristors are fired in the sequence of their number with a phase difference of 60^0 by gate pulses of 120^0 duration.

-Each thyristor conducts for 120^0 and two thyristors conduct at time, one from the upper group (odd numbers) and the other from the lower group (even numbers) applying respective line voltage to the motor.

-Transfer off current from an outgoing to incoming thyristor can take place when respective (المعني) line voltage is of such a polarity that not only if forward biases the incoming thyristor, but also leads to the reverse biasing of the outgoing when incoming turns- on.

- Thus, firing angle for a thyristor is measuring from the instant when respective line voltage is zero and increasing.

- Hence, for thrirystor T1, firing angle (α) is measured from the instant Vac=0 and increases.

-If line voltage is taken as the reference voltage, then $V_{ab}=V_m \cos \omega t$, $\alpha=\omega t-(\pi/3)$ where V_m is the peak line voltage.

- the discontinuous conduction is neglected, because it occurs at a narrow region of its operation

Example: A 220V, 1500 rpm, 50A separately excited dc motor has an armature resistance of 0.5 Ohm, is fed from a 3-phase fully- controlled rectifier. Available ac source has line voltage of 440V, 50Hz. A star/delta connected transformer is used to feed the armature with 220V when the firing angle is zero.
Calculate the transformer turns ratio.
Determine the value of firing angle when:
a) Motor is running at 1200rpm and rated torque.
b) Motor is running at -800rpm and twice the rated torque.

Solution : The peak value of transformer output at motor nominal voltage is $V_m = \frac{\pi}{3}\frac{V_a}{cos\alpha} = \frac{\pi}{3}\frac{220}{cos0^o} = 230.4V$

rms converter input voltage between lines = $\frac{230.4}{\sqrt{2}} = 162.9V$

For star-delta transformer connection, ratio of turns between phase windings of primary and secondary = $\frac{440/\sqrt{3}}{162.9} = 1.6$

at 1500rpm E=220-0.5×50 = 195V

at 1200rpm $E = \frac{1200}{1500} \times 195 = 156V$ Va =E+IaRa=156+50×0.5=181V

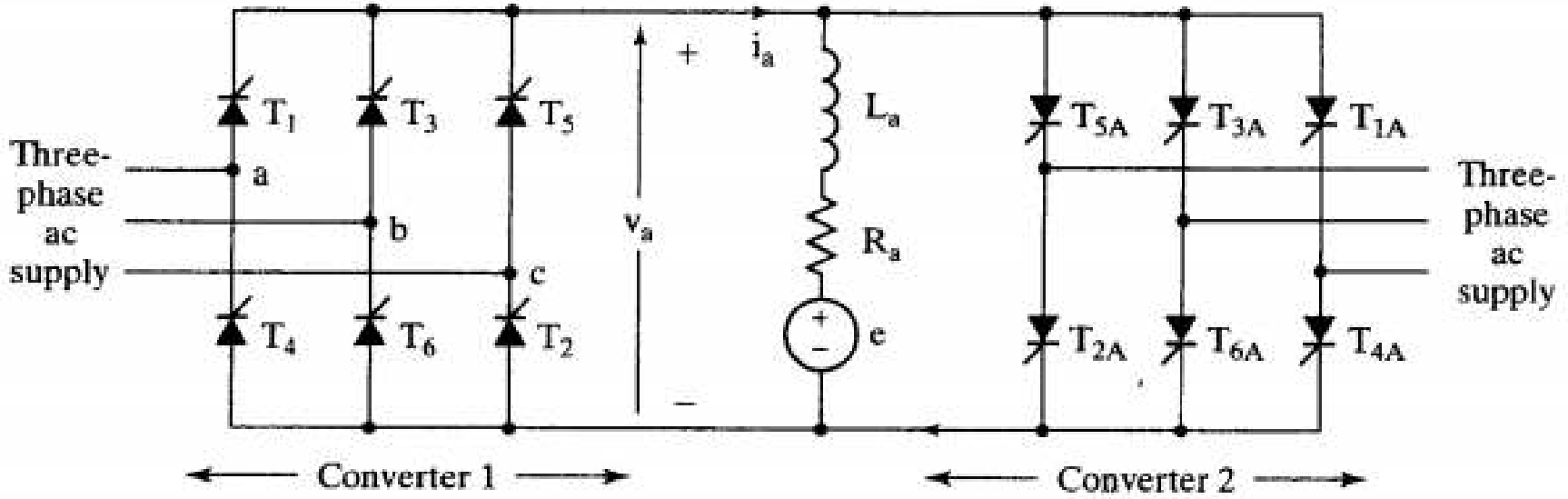
Since $V_a = \frac{3V_m}{\pi} cos\alpha$, $cos\alpha = \frac{\pi}{3}\frac{V_a}{V_m} = \frac{\pi}{3}\frac{181}{230.4} = 0.8227 \Rightarrow \alpha = 34.65^o$

at -800rpm $E = \frac{-800}{1500} \times 195 = -104V$, Va =E+IaRa=-104+100×0.5 = -54V

$$cos\alpha = \frac{\pi}{3}\frac{V_a}{V_m} = \frac{\pi}{3}\frac{-54}{230.4} = -0.245 \Rightarrow \alpha = 104.20^o$$

Three-Phase Dual-Converter Drives

- Two three-phase dual- converters are connected in an arrangement similar to in the single -phase
- Either converter 1 operates to supply a positive armature voltage, V_a or converter 2 operates to supply a negative armature voltage, $-V_a$.
- It is a four-quadrant drive and is limited to applications up to 1500 kW.
- The field converter can be a full-wave converter.



Three-Phase Dual-Converter Drives

If converter 1 operates with firing angle of α_1 , the average armature voltage is

$$V_a = \frac{3\sqrt{3}V_m}{\pi}cos\alpha_1 \quad \text{for} \quad 0 \le \alpha_1 \le \pi$$

Where V_m is a peak value of phase voltage of a Y- connected three-phase ac supply.

If converter 2 operates with firing angle of α_2 , the average armature voltage is

$$V_a = \frac{3\sqrt{3}V_m}{\pi}cos\alpha_2 \quad \text{for} \quad 0 \le \alpha_2 \le \pi$$

With three-phase full converter in the field circuit, the average field voltage is

$$V_f = \frac{3\sqrt{3}V_m}{\pi}cos\alpha_f \quad \text{for} \quad 0 \le \alpha_f \le \pi$$

Resume: for continuous current of sep. excited dc motor drive

Converter (rectifier)	Single-phase Half-wave	Single-phase full-wave semi(half)-controlled	Single-phase full-wave fully-controlled
Armature voltage (Va)	$\frac{V_m}{2\pi}(1+cos\alpha)$ $=0.225V_s\ (1+cos\alpha)$	$\frac{V_m}{\pi}(1+cos\alpha)$ $=0.45V_s(1+\cos\alpha)$	$\frac{2V_m}{\pi}cos\alpha$ $=0.9V_s cos\alpha$

Converter (rectifier)	Three-phase half-wave	Three-phase full-wave semi (half)-controlled	Three-phase full-wave fully – controlled
Armature voltage (Va)	$\frac{3\sqrt{3}V_m}{2\pi}cos\alpha$ $=1.17V_s cos\alpha$	$\frac{3\sqrt{3}V_m}{2\pi}(1+cos\alpha)$ $=1.17V_s\ (1+cos\alpha)$	$\frac{3\sqrt{3}V_m}{\pi}cos\alpha$ $=2.34V_s cos\alpha$

$V_m = \sqrt{2}V_s = \sqrt{2}V_{s\,rms}$- peak value of the supply phase voltage,

peak value of the line voltage $V_{Lm} = \sqrt{3}V_m$